

ZELIHA URAL MERPEZ

Machine Learning Engineer · Applied AI R&D

Richmond, BC, Canada · zelihamerpez@gmail.com · (778) 939-7183

PhD, Mathematics — Bilkent University · MSc, Data Science — University of British Columbia

PROFESSIONAL SUMMARY

Machine Learning Engineer with a PhD in Mathematics and an MSc in Data Science, specializing in applied AI R&D: training, fine-tuning, evaluating, and deploying deep learning and transformer-based models under real-world constraints. Currently leading an end-to-end computer vision program from research prototype through on-device edge deployment (TensorRT/ONNX, containerized multi-model inference) for offline, connectivity-constrained environments. Hands-on experience evaluating and benchmarking LLM outputs, building Unity/C# real-time simulations, and turning novel ideas into working demos, pipelines, and full-stack products.

TECHNICAL SKILLS

Languages: Python, C#, SQL, JavaScript / TypeScript, R, Dart

Machine Learning: Deep learning (PyTorch), model fine-tuning, transformer-based models, classification & detection, computer vision, hyperparameter optimization, model evaluation & validation, statistical & predictive modelling

On-Device & Edge AI: TensorRT runtime, ONNX, Docker multi-model orchestration with priority scheduling, offline-first inference, on-device data plane (preprocessing, filtering, aggregation with cloud sync), version-controlled model registry, hybrid NB-IoT / LTE-M (NTN) connectivity

LLMs & Evaluation: Generative-model evaluation & ranking, benchmark authoring, factuality & reasoning assessment, A/B testing & experiment design, large-scale data collection

Simulation & Apps: Unity (C#), real-time simulation, Streamlit dashboards, Next.js, REST APIs, Plotly / Matplotlib / Seaborn, data storytelling

Platforms & Tools: AWS (SageMaker, Cognito), GCP (Cloud Run, Firebase), MongoDB, PostgreSQL, pandas, NumPy, scikit-learn, Git, Jupyter, MkDocs, Netlify, Flutter (cross-platform iOS/Android)

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | M2M Tech Inc. *Dec 2024 – Present*

- Lead the DeckGuardVision computer vision program, overseeing engineering work, technical direction, and delivery across the team.
- Develop and optimize deep learning models for computer vision applications on large-scale image datasets; fine-tune pretrained architectures and lift model performance through systematic experimentation and evaluation.
- Architect on-device inference for edge hardware deployed overseas without reliable connectivity: TensorRT/ONNX runtimes, Docker + K3s multi-model orchestration, and priority scheduling across models on a single device.
- Automate training and inference pipelines to improve reproducibility and deployment efficiency; design interactive dashboards for monitoring model predictions and performance metrics.
- Build full-stack web applications in Next.js, integrating APIs, databases, and backend services.

AI Development Specialist / AI Math Analyst | Outlier *Mar 2024 – May 2025*

- Evaluated and ranked generative-AI (LLM) model outputs across mathematics and computer science domains, assessing factuality, reasoning quality, and consistency.
- Authored complex benchmark questions and delivered detailed, data-driven feedback to guide model-improvement and fine-tuning initiatives.

Postdoctoral Researcher | University of British Columbia *Sep 2021 – Oct 2023*

- Built predictive and statistical models for stock-price forecasting, information-risk analysis, and quantitative financial decision-making, applying time-series methods to U.S. market data.
- Applied large language models and large-scale data-collection techniques to business-intelligence research; built automated ETL pipelines transforming large Excel datasets into structured relational databases.

Database Technology Course Instructor | University of British Columbia *Sep 2021 – Dec 2022*

- Designed and taught course material on database design, normalization, SQL, and modern data storage architectures (data lakes, warehouses, relational and NoSQL databases), with practical, real-world assignments.
- Delivered the course to both undergraduate and MBAN cohorts, adapting content from foundational SQL to business-focused analytics use cases for each audience.

SELECTED PROJECTS

DeckGuardVision — Edge-Deployed Safety Violation Detection — M2M Tech · Project Lead · Current

- Lead the project end-to-end and oversee work delivered by the team, from research and model development through documentation and deployment.
- Built an end-to-end computer vision pipeline for zone-based safety-violation detection in surveillance video, including an interactive zone-definition tool and an automated labeling workflow.
- Fine-tuned person-detection models and benchmarked state-of-the-art approaches across PPE, safety-zone violation, and fall-detection domains.
- Driving edge deployment to overseas sites that must operate offline: AI inference via TensorRT runtime with ONNX/PyTorch/TensorFlow model support, orchestrated with Docker for multi-model runtime.
- Designed the local data plane — on-device preprocessing, filtering, and aggregation with cloud sync when connectivity returns — over hybrid NB-IoT / LTE-M (NTN) connectivity with dual-SIM cellular failover for remote deployments.
- Authored project documentation in MkDocs and deployed it via Netlify.

Stack: Python, PyTorch, TensorRT, ONNX, Docker, AWS SageMaker, MkDocs, Netlify

AgriTwin — Mushroom Growth Simulation — M2M Tech

- Built a Unity-based real-time simulation platform modeling mushroom growth across multiple species, with image-based initialization and environment-aware growth-parameter tuning.

Stack: Unity, C#, Python

StockVision — Agent-Based Stock Trading Simulation — University of British Columbia (Postdoctoral Research)

- Designed and implemented autonomous trading agents with distinct decision-making strategies (adaptive vs. flexible, fast vs. slow), each maintaining per-stock portfolio weights, adaptive wait times, and performance history that dynamically shifts capital toward better-performing assets.
- Backtested agents over 5 years of historical market data (10 stocks, ~1,200 trading days) using moving-average crossover signals, tracking daily portfolio value, per-stock gains, and transaction behavior.
- Optimized agent decision thresholds via grid search (800 configurations) and modeled realistic transaction-cost constraints, comparing strategy outcomes through portfolio simulations.

Stack: Python, pandas, NumPy, scikit-learn, Matplotlib, yfinance

Earnings Call Text Analytics & Market Reaction Study — University of British Columbia (Postdoctoral Research)

- Extended the methodology of the "Number of Numbers" research paper by building an NLP pipeline with Microsoft Recognizers-Text to extract and classify numerical mentions in earnings call transcripts with higher precision — separating currency, percentage, ordinal, and plain-number categories and correctly handling linguistic modifiers such as "negative."
- Engineered rhetorical features quantifying ethos, pathos, and logos usage in executive language during earnings calls.
- Applied event-study statistical methods to link numerical density and rhetorical features to cumulative abnormal returns (CAR), measuring how communication style in earnings calls relates to market reaction.
- Measured market reaction across multiple horizons — short-term CAR in the [-1, +1] event window, plus post-earnings-announcement drift over 1-, 2-, and 3-month windows — to distinguish immediate market response from longer-term price adjustment.

Stack: Python, Microsoft Recognizers-Text, NLP, pandas, statistical modeling, event-study methodology

BF & PropertyGuys — Full-Stack Web Applications — M2M Tech

- Developed and maintained frontend and backend features for production web applications, focusing on user authentication flows and database management.
- Deployed and served both frontend and backend on AWS ECS (containerized), routing traffic through AWS API Gateway; managed user authentication and data with Firebase, AWS Cognito, and MongoDB.
- Integrated the Repliers real-estate API with Mapbox for interactive, map-based property search; supported deployment and testing of the cross-platform Flutter mobile app.

Stack: Next.js, Node.js, Flutter, Firebase, AWS Cognito, MongoDB, Mapbox, REST APIs

EDUCATION

PhD, Mathematics | Bilkent University

MSc, Data Science | University of British Columbia